

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

A sad farewell to Kepler, discovering the dust moons of Earth, vaping can be good for you, how language changes colour perception, kids and the effect of screen time and the Parker Solar Probe



X-Raying the Enigma Machine

A team of researchers have explored the interiors of the Enigma Machine, used by the Germans during WWII to send coded messages. X-Raying the machine allows the researchers to explore the individual wires and pins within the device, without destroying it. This is the first time that scientists have got such a close look at the innards of an Enigma Machine.

<https://digit.in/enigmaxray>

WHAT'S NEW

Windmills are an alpha predator

Wind farms are considered environmentally friendly option for generating electricity. About 8 percent of the power in India is generated using wind. Researchers from the Indian Institute of Science, studying the ecological impact of the wind farms found that there are far reaching consequences of these windmills. In the same plateau in the Western Ghats, researchers studied predatory birds such as hawks and kites, at locations with and without wind farms. Predatory birds were far fewer in the locations with wind farms. Also, birds in the vicinity of wind farms made fewer attempts at capturing prey, and displayed a reduced frequency of dives. Consequently, there was an increase in the



number of the prey species of these birds, the fan throated lizards. What is happening here is known as a trophic cascade. Widespread changes occur in the food pyramid when a predator species is introduced. The windmills are essentially playing the role of a new apex predator in the region, disrupting the balance in the area.

<https://digit.in/predwm>

Farewell to Kepler

The work on Kepler actually began in 1983. The technique used by Kepler to detect exoplanets is known as "Transit Photometry". When a planet passes in front of a star, there is a slight dip in the brightness of the star. By observing the dips in the brightness over time, a telescope could measure the frequency and intensity of the dips. These measurements, could, in turn, tell the orbital period and size of the planets. After years



of improvements, NASA greenlit the Kepler Space Telescope in 2001. It took another 8 years to build and deploy, and began its operations by simultaneously monitoring 170,000 stars for dips in brightness,

in the constellations of Lyra and Cygnus. Within a year of its launch, Kepler had identified five exoplanets. Now, nine and a half years later, Kepler has identified a staggering confirmed 2,662 exoplanets! An additional 2,244 exoplanet candidates have also been identified. Kepler has studied 530,506 stars, collected 678 GB of data, which has been used to publish 2,946 scientific papers! <https://digit.in/keplerbyebye>



Modes of the brain

The prefrontal cortex of the brain stores multiple rules to use in different scenarios. When you drive slowly in a residential area with children playing, or when you tiptoe into a library, these rule sets kick in. Research into this cognitive flexibility can allow developers to improve artificial intelligence.

<https://digit.in/brainmodes>



The crisis of science

Studies in medicine, psychology, economics and other fields are increasingly showing different results when replicated. This is cause for concern, and questions the validity of these fields. Now, a pair of new studies explores the replication crisis to try and find solutions.

<https://digit.in/scicrisis>



Traffic frogs

Frogs, like humans, are stressed by the noise of traffic. A study shows that frogs living in ponds closer to highways have dampened stress and immune responses to traffic, while frogs living in isolated ponds are more easily affected. Researchers want to know how they adapted.

<https://digit.in/frogstress>